

Math 1314 Lesson 06

1) Solve the equation $4x^2(x^2 - 16) = 0$

a) Set $4x^2 = 0$ and solve for x . $x = \underline{\hspace{1cm}}?$

b) Set $x^2 - 16 = 0$ and solve for x . $x = \underline{\hspace{1cm}}?$

c) Graph the function $y = 4x^2(x^2 - 16)$. x -intercepts are $\underline{\hspace{1cm}}?$.

2) Solve the equation $2x^4 - 10x^2 = 0$

a) Factor $2x^4 - 10x^2$. $2x^4 - 10x^2 = 2x^2 \cdot (\underline{\hspace{1cm}}? \text{ missing factor})$

b) Set $2x^2 = 0$ and solve for x . $x = \underline{\hspace{1cm}}?$

c) Set missing factor from part (a) equal to zero and solve for x . $x = \underline{\hspace{1cm}}?$

d) Graph the function $y = 2x^4 - 10x^2$. x -intercepts are $\underline{\hspace{1cm}}?$

3) Solve the equation $12x^2 = 36x^4$

Hint: Subtract $36x^4$ from both sides first.

a) Factor $12x^2 - 36x^4$. $12x^2 - 36x^4 = 12x^2 \cdot (\text{missing factor ?})$

b) Set $12x^2 = 0$ and solve for x . $x = \underline{\hspace{1cm}} ?$

c) Set missing factor from part (a) equal to zero and solve for x . $x = \underline{\hspace{1cm}} ?$

d) Graph the function $y = 12x^2 - 36x^4$. x -intercepts are $\underline{\hspace{1cm}} ?$

4) Solve the equation $5x^3 = 25x^5$

Hint: Subtract $25x^5$ from both sides first.

a) Factor $5x^3 - 25x^5$. $5x^3 - 25x^5 = 5x^3 \cdot (\text{missing factor ?})$

b) Set $5x^3 = 0$ and solve for x . $x = \underline{\hspace{1cm}}?$

c) Set missing factor from part (a) equal to zero and solve for x . $x = \underline{\hspace{1cm}}?$

d) Graph the function $y = 5x^3 - 25x^5$. x -intercepts are $\underline{\hspace{1cm}}?$

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